

**Embedded Systems Project**

**DESIGN REPORT #2**

**Title: Technical Characterization**

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# 1. Introduction

This aim of this project is to develop an autonomous buggy capable of travelling around a track, turning in a controlled manner and coming to a controlled stop. The production consists of two aspects, the mechanical aspects, ensuring it is capable of meeting the physical requirements, and the technical aspects, ensuring that it receives the information and understands how to process it in order to govern itself. Design report 1 covered the choice of one of the main mechanical components chosen, the motor, this report delves into the technical aspects, which will determine how the motor system is communicated with, the algorithms used to control it and a justification for the chosen sensors.

The environment in which our buggy operates will pose a major challenge in the execution of this project, our buggy will be exposed to a spectrum of wavelengths at any point throughout navigation, which can saturate detectors and complicate comparator circuit logic. In order to resolve this problem a sensor must be selected that avoids the wavelengths present in said light sources, as well as remains efficient and functional in bright sunlight. Therefore the current through the TCRT5000 must be measured with varying background illumination in order to gauge it's performance in said conditions, which will be covered in section 3. During this report, a chassis design will be suggested, a key consideration during the design process is the optimal height at which the TCRT5000 sensor is positioned. This height is not arbitrary; it is a result of several tests carried out on the sensor, which will be referenced as justification for the chassis design in section 7.

Two classifications of sensors will be used during this project, non-line and line sensors, each with distinct functionalities, a goal of this report is to clarify the unique role both play in the overall system and the necessity for them both. The most obvious need for non-line sensors is determining the speed and direction of the motors. A quadrature encoder will be in either motor, they generate two square wave signals (A and B channels) with a phase difference of 90 degrees. This phase information allows the system to determine both the direction and the amount of rotation on each motor. In addition to this a current sensing circuit to monitor the potential difference through the motors. The signals from the non-line sensors will be continuously integrated into a closed-loop feedback system alongside the signals from the line sensors, to inform the control algorithm how to adjust the movement of the buggy accordingly.

How the external stimuli is incorporated into the system will be represented through the use of context and use case diagrams, highlighting how each element interface and communicates with one another. The interaction that leads to movement is from the microcontroller, through the GPIO pin, to the motor drive board. The main function of this board is to amplify the motor output pins from the STM32, since it can only drive a maximum current of around 20mA, which wouldn't suffice for bringing our motor to top speeds and up inclines, whereas the drive board can reach up to 1.4 A. The STM32 provides a digital signal out, whereas the motors require an analogue signal, therefore a PWM signal is used to approximate the analogue signal. The microcontroller will vary the duty cycle of the PWM signal depending on calculations carried out in the control algorithm, and a low pass filter will be applied to ensure an average DC is sent to the motor.

## 2. Software

### 2.1 Functional Summary

The buggy is situated at the start of the white line and turned on, the software will begin to run and the system will respond to and process external stimuli to detect and follow the white line, autonomously navigating the buggy around the track without touching the barriers nor stopping at line breaks. Depending on the feedback from the sensors, each motors speed and direction will be changed in accordance. It should stop at the end of the line and wait to detect a BLE signal, which will trigger the buggy to return to the start of the track, where, once reached, should come to a controlled stop with all parts within 20 cm of the end of the track.

### 2.2 Software constraints

Whilst creating such a system there are several encumbrances, both in the physical and digital which can lead to constraints in the production of our software, it is important to acknowledge both and recognize how they will affect our system.

1. Memory
2. Real-time constraints (Sampling frequency)
3. Clock frequency
4. Power consumption
5. Environmental Constraints

For this project an STM32F401RE Nucleo board will be utilised. This board is equipped with 512-Kbytes (equivalent to 524,288 bytes) of flash memory, which is primarily used for storing the compiled program. Considering the 1:1 ratio of characters to bytes, adhering to the industry standard of a maximum of 80 bytes per line of code, a theoretical maximum of 6,553 lines can be derived. This is assuming that all flash memory would be used for the code, which may not be the case, however an assumption taken is that it won't need to store run-time variables as the STM32 board has 96 Kbytes of SRAM available to do this task. It is also unlikely that this project will require 6,553 lines of code, however this estimate gives head room for external libraries and dependencies used within the system, also for the varying size of the compiled code. Therefore the memory provided by the microcontroller won't be a limitation on this project.

How often the microcontroller executes the control loop plays a critical role in how fast the buggy's response time is, the system is time sensitive and it's therefore critical that instructions are being run in due timing. The sample frequency of the control loop needs to be faster than the fastest response time required during the navigation of the buggy. It's known that each wall will have a distance of 140mm from either side of the white line, and the white line has a width of  $17\text{ mm} \pm 3\text{ mm}$ , assuming it to be 14 mm in order to obtain the shortest time needed. At each point in a turn there will be a distance of  $140+14 = 154\text{ mm}$  from the centre of the buggy to the nearest wall. From the last design report, it was found that the theoretical top speed of the buggy was about 6 m/s, therefore the response time required to alter the direction of the buggy and prevent it hitting the wall will be:  $0.154/6 = 0.0257\text{ s}$  or 25 ms. This response time constraint should be taken into account when adjusting the sampling frequency for the control loops, as it directly influences the system's ability to respond effectively.

The clock frequency represents how fast the STM32's CPU processes instructions, the faster the frequency, the more clock cycles per second and therefore the more instructions executed per second. Since the ARM®32-bit Cortex®-M4 CPU has a clock frequency of up to 84MHz, it can execute  $85 \times 10^6$  clock cycles per second, therefore, in 0.25 ms (required response time), it would perform approximately 21,250 clock cycles. Though it doesn't change the sampling frequency, this calculation shows that there is ample processing power during the sampling period. This may be sufficient but since the response time is such a short period, it's paramount that the code is written as efficiently as possible to ensure it is executed as fast as possible. High clock frequencies are wanted, however utilising them can result in increased power consumption and heat generation, which is not ideal. It's once again critical that the frequency constraint is recognised, and the code is made as efficient as possible.

To touch on the power consumption constraint, the power source needs to be taken into consideration, the autonomous buggy will strictly operate off of a finite 10V power source, the battery packs, it's in best interest of the project to utilise the power provided as effectively as possible in order to maximise the health and operation of the batteries. MBED offers 3 power modes for power optimisation: Active, Sleep, and Deep sleep. Switching between these modes occurs automatically. When all threads in the system are idle, the idle thread invokes the sleep manager to bring the system to a lower power consumption mode. This means that code which is reliable and executes tasks faster, will allow the MBED OS to trigger the microcontroller to spend more time in low-power modes, in comparison to inefficient loops and code which will take longer to execute, not only reducing the microcontrollers time in lower power modes, but also directly requiring more processing power than cleaner code. It's therefore paramount that the logic integrated in the software is as well written as possible and making the most use of our finite power source.

It is also essential that consideration is taken when choosing the switching frequency for the PWM signal within the code, the switching frequency is the rate at which the DC voltage is switched on and off during the pulse width modulation. The buggy needs to operate smoothly and without ripple so the switching frequency needs to be suitably fast otherwise the system would suffer from higher conduction losses, however switching more often than needed leads to higher switching losses since the H-bridges are semiconducting devices which dissipate energy when transitioning. It's therefore critical that within the codebase the switch frequency is acknowledged in order to adhere to the power constraint. The mode of operation for the motors is also a critical factor since each mode provides their own advantages, for example bipolar mode allows energy recovery during braking, since the motor acts as generator, but it can also be less efficient if going from full speed to full speed in opposite directions, since a H-bridge is required to reverse current directions. On the other hand unipolar provides a dedicated pin for direction control, simplifying the system and reducing losses when turning. The choice of mode needs to be considered whilst coding since it will have an effect on the power losses in the system.

The main sensor for line detection is the TCRT5000, it has an operating temperature of -25°C to +85°C, and it is therefore unlikely that the temperature of our environment will have any effect on the efficiency and accuracy of said sensor, however there are more environmental attributes to consider. The track will have segments of darkness, brightness and luminous lights with varying wavelengths, under such conditions our TCRT5000 will have varying difficulty identifying the white line and thus have a

varying sensitivity to its presence. This will need to be accounted for, as if a low magnitude response from the TCRT5000 (Comparing to when there is a fully lit environment) is interpreted as the absence of a white line, logic, which for example may cause the buggy to turn, may be implemented, when in reality the low magnitude signal may mean the white line is fully present, but the environment is darker. The environmental constraints will therefore need consideration and affect the way our code is written.

## 2.3 Context diagram

Sensors for motion

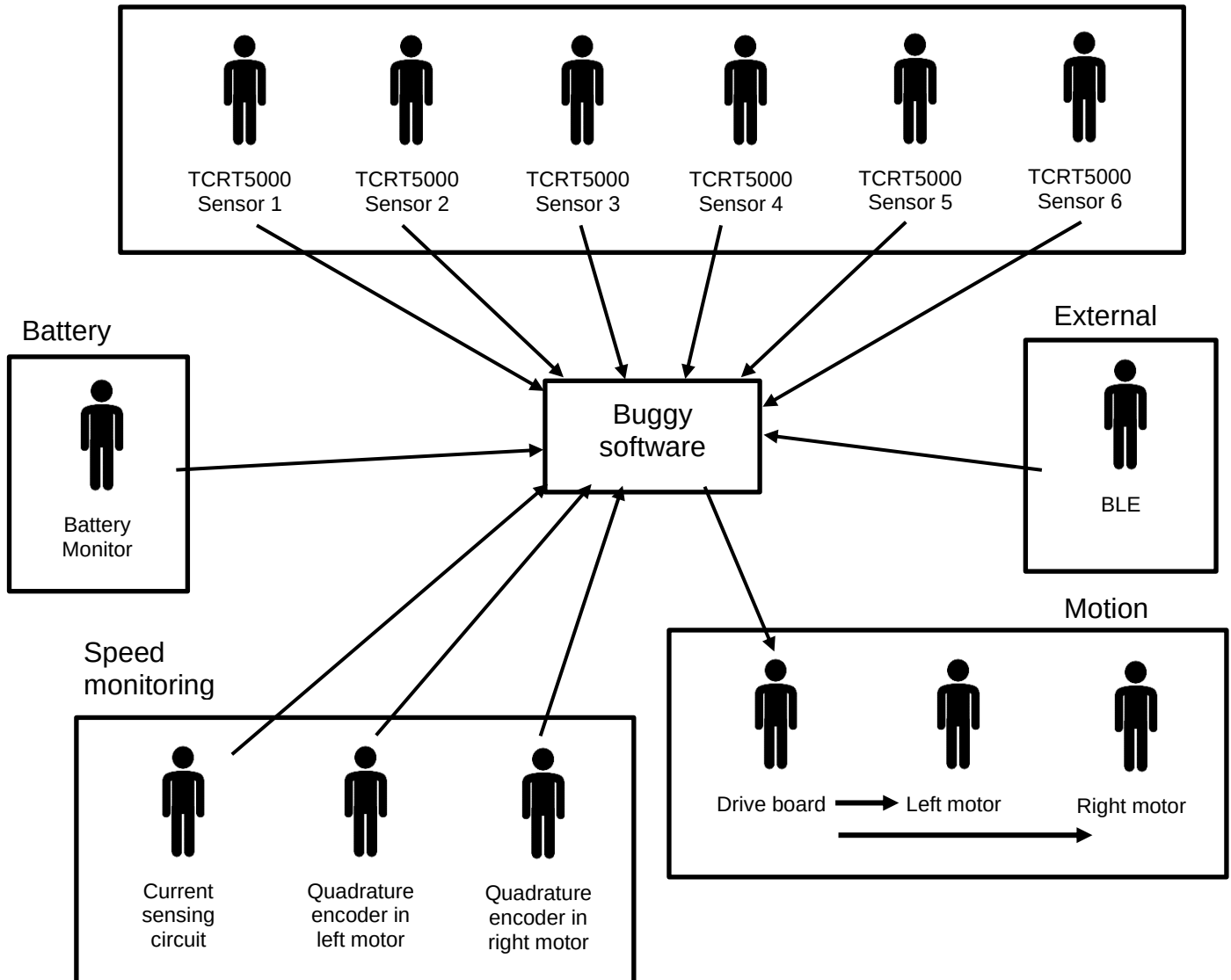


Figure 2.3

The buggy's software system is represented by the black box in the middle, around it are stick people, which represent elements of the wider system. The context diagram provides an abstract visualisation of the interaction between the buggy's software and these peripheral elements, in particular the direction in which data flows between them.

## 2.4 Table of messages

Table 2.4

| Source                  | Destination       | Interface             | Arrival pattern | Size    |
|-------------------------|-------------------|-----------------------|-----------------|---------|
| BLE signal              | Buggy software    | GPIO PWM              | Aperiodic       | TBD     |
| TCRT5000                | Buggy software    | GPIO                  | Periodic        | 32 bits |
| Quadrature encoders     | Buggy software    | GPIO                  | Periodic        | 32 bits |
| Battery monitor         | Buggy software    | GPIn 8 of motor drive | Periodic        | TBD     |
| Current sensing circuit | Buggy software    | GPIO                  | Periodic        | TBD     |
| Motor drive board       | Motors            | GPIO PWM              | Continuous      | TBD     |
| Buggy software          | Motor drive board | GPIO                  | Aperiodic       | TBD     |

## 2.5 Use case description

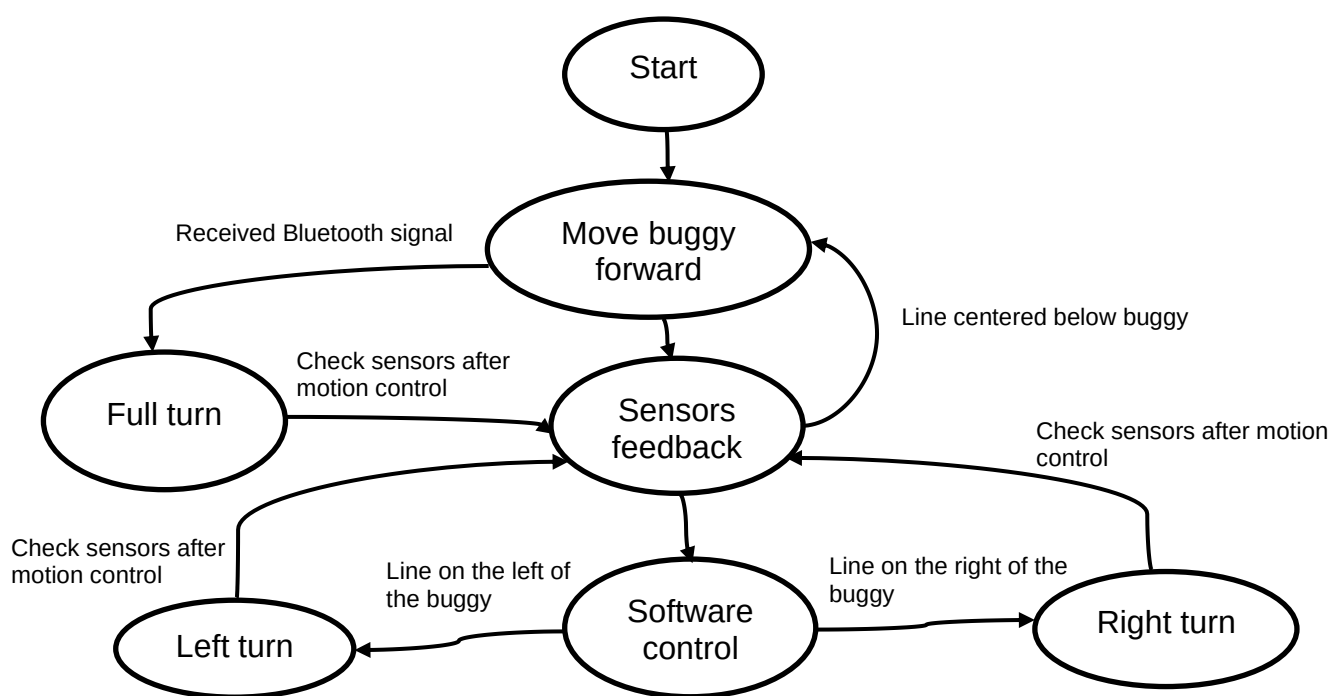
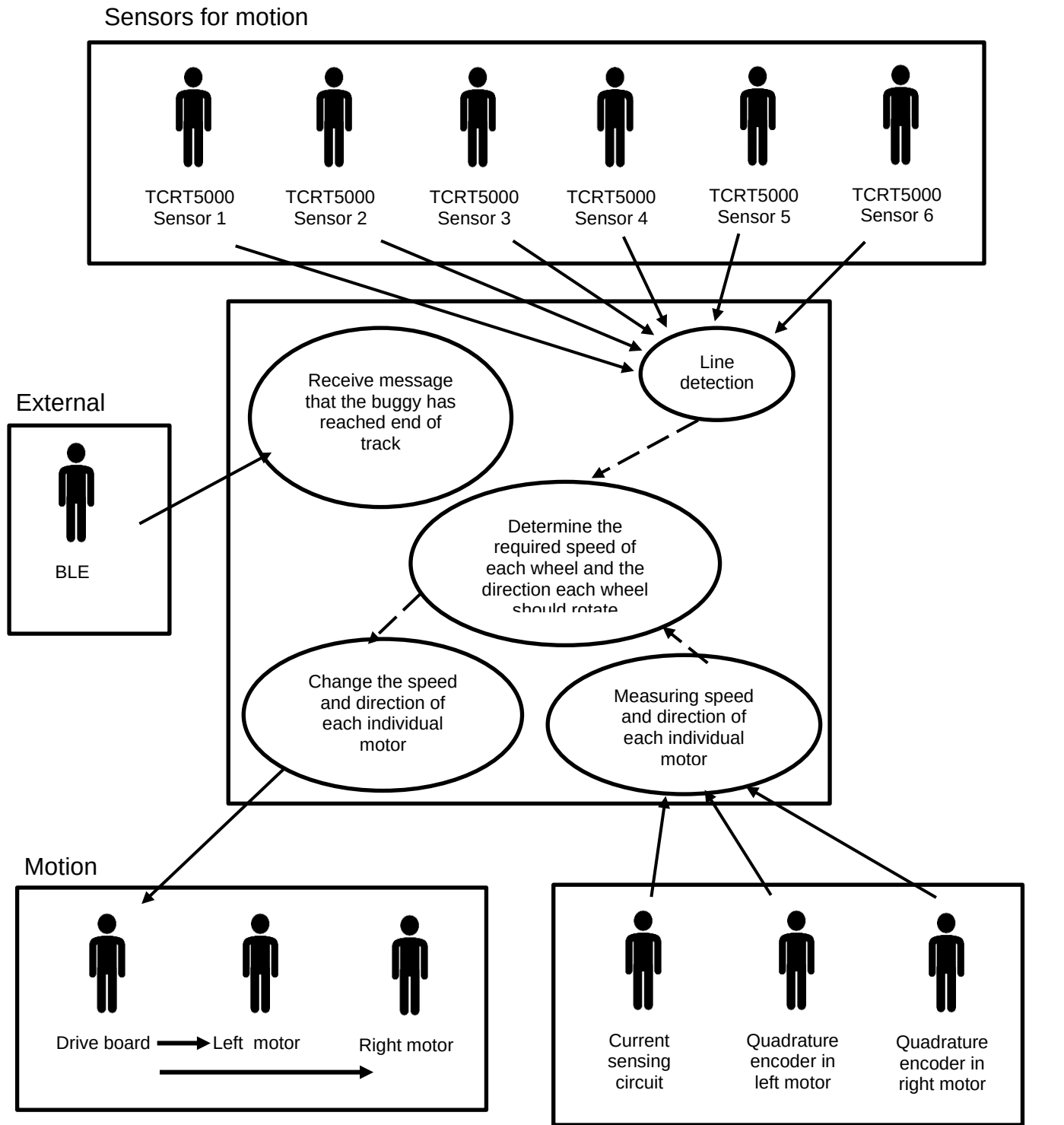


Figure 2.5

## 2.6 Use case diagrams



This use case diagram describes the main functions of the software, the inputs required for the functions and finally where the output of these functions will end up, giving an overview of the system workflow.

## 2.6 Object specifications

Code 2.6



```

//Motion Control
void leftTurn(void);
void rightTurn(void);
void forward(void);

//Sensors
float sensor1(void); //Function to recieve sensor1 data
float sensor2(void); //Function to recieve sensor2 data
float sensor3(void); //Function to recieve sensor3 data
float sensor4(void); //Function to recieve sensor4 data
float sensor5(void); //Function to recieve sensor5 data
float sensor6(void); //Function to recieve sensor6 data

//Wheel Speed
float wheel_velocity_helper(float current_encoder_ticks, float previous_encoder_ticks, float sample_time); //Function
to calculate wheel velocity using encoder data
float wheel_velocity_left, wheel_velocity_right; //Wheel velocity
float robot_angular_velocity_helper(float wheel_velocity_right, float wheel_velocity_left); //Function to calculate
robot angular velocity using wheel velocity
float robot_angular_velocity;
float current_sensing_power_levels(void); //Get the power levels from the current sensing circuit

//Bluetooth
void recieve_bluetooth_data(void); //Function to recieve bluetooth data
float bluetooth_data; //Bluetooth data

//Battery Monitoring
float get_raw_voltage_reading(void); // Function to get raw voltage
float get_raw_current_reading(void); // Function to get raw current
int voltage_reading_raw, current_reading_raw; // Variables to store raw voltage and current readings
float calculate_battery_voltage(int voltage_reading_raw); // Function to calculate the voltage of the battery given the
raw voltage reading
float calculate_battery_current(int current_reading_raw); // Function to calculate battery accumulated current given
the raw current reading
float Voltage, Current; // Variables to store processed battery voltage and current

```

### 3. Line Sensor Characterisation

During the laboratory a series of experiments were performed on a combination of sensors in order to determine the sensor most suited for the task of detecting the white line under circumstances similar to those on race day. The category of sensors tested are considered infrared sensors, in this case a combination of an infrared LED and a phototransistor. These sensors emit infrared light and measure the intensity of the reflected light, IR sensors were chosen over visible light sensors as they are affected less by background illumination, providing more consistent results in the detection of the white line. When light is received by a phototransistor it will induce a current flow in the semiconductor material, which will be amplified, and the amount of current through the phototransistor is proportional to the intensity of incident light, meaning the light and dark surfaces can be differentiated.

The first sensor tested is the TCRT5000, followed by the combinations of the TSHA4401 infrared LED paired with BPW17N phototransistor, and TSHG6400 infrared LED paired with TEKT5400S phototransistor. The production of the IR sensors required the LED-phototransistor pair to be soldered to a printed circuit board, taking consideration of polarities. In order to obtain accurate results, it's necessary to prevent crosstalk within the sensor, therefore the next step was to ensure there's a shield between the emitter and detector using opaque plastic superglued to the PCB, to ensure the light detected from the emitter is only reflected light. After these two 4-pin male headers were soldered on either side of the PCB to allow this configuration to be slotted into a breadboard connected a MyDAQ. Once the sensor was constructed its circuit had to be determined, the datasheets of the individual components were referenced in order to find the respective resistor values required to produce the ideal current through them. This process took several iterations to find the ideal resistors for our particular use case, which validated the

decision to initially use a breadboard as it allows for modularity and ease in configuration, once resistor values were decided upon the circuit was created on a stripboard which could be placed into a test rig with slots of varying height, ready for experiments to be carried out.

## **3.1 - Experimental methods**

### **3.1.1 - Constructing the circuit**

The applied voltage to the sensor bracket is set at 5 V, while the infrared LEDs operate at around 1.2 V. Through careful consideration, a 330-ohm resistor has been determined as the optimal choice. This selection allows for a current of 11.5 mA, a value sufficiently close to the ideal 10 mA specified in the infrared LEDs' datasheets. The decision to opt for a slightly lower resistance aims to achieve a slight increase in LED brightness.

As for the resistor connected to the phototransistor, the underlying assumption is that the phototransistor behaves as a closed switch when exposed to light. In line with this assumption, a 3000-ohm resistor is employed. This choice ensures sensitivity to even subtle changes in voltage across the phototransistor, enhancing its capability to detect variations in light levels.

## **3.2 - Choosing the line sensors**

For this lab, 3 sensors were successfully created and tested:

1. TCRT5000
2. TSHA4401 infrared LED paired with BPW17N phototransistor
3. TSHG6400 infrared LED paired with TEKT5400S phototransistor

### **3.2.2 - Pairing the components of the sensors**

The combination of emitter and receiver sensors was decided based on the wavelength of light emitted by the LED alongside the range of frequencies that are detectable by the phototransistors. The datasheets for each sensor were utilised to create the characteristic table 3.2.3. From this table it is known that the TSHA4401 infrared LED emits electromagnetic waves at a peak wavelength of 875 nm, and the BPW17N is capable of detecting wavelengths from 620nm to 960nm, with a peak sensitivity at 825 nm, meaning that it is able to optimally detect the wavelengths of light coming from the TSHA4401 LED. The TSHG6400 LED emits a peak wavelength of 850 nm, and the TEKT5400S phototransistor has a detectable range from 850 nm to 980 nm, making it suitable for the detection of the emitted light.

A visible light sensor has been created by pairing the OVL5521 LED and the SFH203P phototransistor, however, since the sensor detected visible light, it was heavily affected by the background illumination and the small gap between the black plastic bit and the PCB. Hence, due to the time constraints of the lab, constructing the sensor again to make it less affected by background light would not have been necessary, since three other sensors have already been constructed and a visible light sensor would not be practical as the light conditions on race day may allow for a high range of inaccuracies depending on how strong the light will be.

### **3.2.3 – Comparing relevant sensor characteristics**

Table 3.2.3 - Sensor Characteristics<sup>[1]</sup>

| sensor | Wavelength | Peak | Half angle | Respons |
|--------|------------|------|------------|---------|
|--------|------------|------|------------|---------|

|                    | range (nm)    | sensitivity (detector) | (Degrees) | e time (μs) |
|--------------------|---------------|------------------------|-----------|-------------|
| TCRT5000           | 950/ N/A      | 940 nm                 | N/A       | N/A         |
| TSHA4401/BPW17N    | 875/(620/960) | 825 nm                 | ±20/±12   | 0.6 / 4.8   |
| TSHG6400/TEKT5400S | 950/(850/980) | 920 nm                 | ±22/±37   | 0.02 / 6    |

### 3.3 - Testing the sensors

In order to figure out which sensor is most suitable for the buggy, a set of tests were performed on each sensor to find out how they will react in different conditions. Four main tests were completed, and results and analysis are shown in the following sections.

#### 3.3.1 – Dark current test

The first test was measuring the dark current of each sensor, to do this the LED of the sensors were turned off and ambient lights were switched off to ensure as little light as possible reached the phototransistors by completely covering them. The current was then measured across the phototransistor. By measuring the dark current when no light is present, the value can be used to calibrate or compensate for this baseline when analysing the signal in the presence of light, ultimately helping improve accuracy of measurements by accounting for the inherent noise introduced by dark current.

Another reason dark current was measured was to determine the base current level that photocurrent must exceed in order to be detected

The results of this test are shown below in table 2:

Table 3.3.1 - Comparing dark current

| Sensor             | Dark Current (nA) |
|--------------------|-------------------|
| TCRT5000           | 2.33333           |
| TSHA4401-BPW17N    | 33.33333          |
| TSHG6400-TEKT5400S | 133.33333         |

Looking at the results from Table 3.3.1, it is obvious that the TCRT5000 sensor has the least value of dark current, meaning that it is able to detect the white lines which much more accuracy than the TSHA4401-BPW17N and the TSHG6400-TEKT5400S.

(93% more accurate than TSHA4401-BPW17N and 98.25% more accurate than TSHG6400-TEKT5400S).

#### 3.3.2 – Background illumination effects

In the second test, the objective was to assess how background illumination affects the current through the sensors, particularly with the goal of achieving a low background illumination current. The method of this test was illuminating the separate phototransistors with a bright torch from equal heights, particularly, 1 metre above the sensor. The reason for desiring a low background illumination current is due to the need for the sensor readings to be predominantly influenced by the light reflected from the white line, this enhances the sensitivity of the line detection also ensuring an accurate and reliable identification. The results of the tests are shown

below:

Table 3.3.2 - effects of background illumination

| Sensor             | Background illumination current (nA) |
|--------------------|--------------------------------------|
| TCRT5000           | 55.66667                             |
| TSHA4401-BPW17N    | 500                                  |
| TSHG6400-TEKT5400S | 1166.66667                           |

The TCRT5000 boast the lowest current due to background illumination, which is an important attribute since it mitigates a constraint previously mentioned in the software section, Section 2. The constraint touched on how environmental factors such as background illumination can affect the accuracy of the sensor and its ability to detect and differentiate between background illumination and the white line. The TCRT5000 mitigates this concern through the use of an in-built filter, which filters out light outside of the infrared spectrum, such as visible light, but since infrared light is still passed through, these results cannot be used to justify the capability of the sensor in direct sunlight, since the sun also emits infrared light. Though the TCRT5000 boasts good results for background illumination current when subject to visible light, it may be affected when there are multiple sources of infrared wavelengths, in this case the sun and those reflected from the white line. A solution to this issue could be using a narrow band filter with a passband that aligns with the wavelength range of the infrared light emitted and reflected by the TCRT5000.

### 3.3.3 – Variation with height

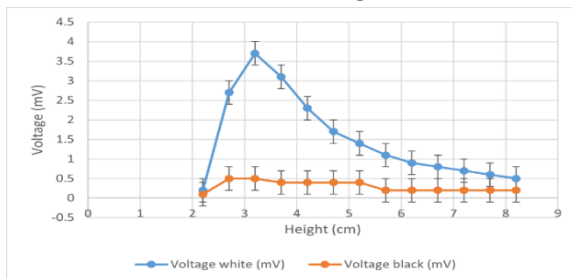


Figure 3.3.3.3: TSHA4401 – BPW17N was measured for both the black and white areas of the tracks. This was done by supplying both the LED and the phototransistor with 5V, situating the created strip board in the plastic rig and recording the voltage across the phototransistor at different heights, ranging from 2.2 cm to 8.2 cm, for both the black area and the white area of the track. The results of the test are shown below:

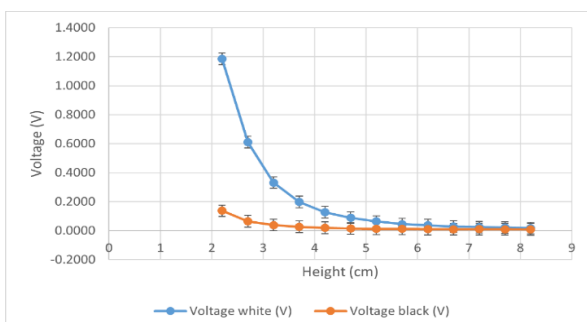


Figure 3.3.3.1: TRCT5000

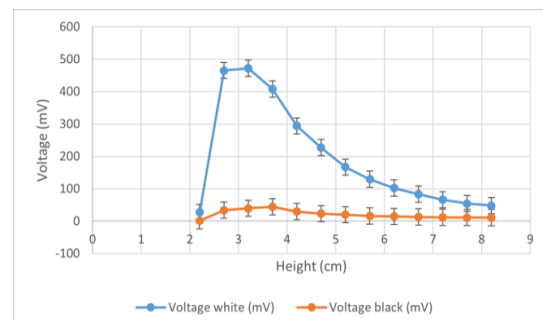


Figure 3.3.3.2: TSHG6400-TEKT5400S

All three sensors respond to the presence of the white line with an increasing voltage with an increasing height from the line, where the closer the sensor is to the white line, the greater the difference between the voltages of the resistor that connects with the phototransistor of the sensor compared to when the sensor is above the black parts of the track. There is, however, an unexpected fall in the graphs of the TSHA4401-BPW17N and TSHG6400-TEKT5400S, and this can be explained due to their longer height compared to the TCRT5000, since at a height of 7.7 cm and 8.2 cm, the black plastic piece that acts as a border between the LED and phototransistor of the sensor gets very close to the track to the point where the light blocked is more than the light reflected.

If the graphs are further analysed, it shows that the y-axis that Figure 3.3.3.1 represents volts, while Figure 3.3.3.2 and Figure 3.3.3.3, the y-axis represents millivolts. This means that there is a much greater difference between the voltage of the resistor of the TCRT5000 when it is on the white line versus when it is on the black part of the track, which allows us to have a smaller margin of error in detecting the white line in relation to the two other sensors, and allows us to have a higher sensitivity. It also showcases the fact that the TSHG6400-TEKT5400S and TSHA4401-BPW17N's have limited effectiveness over different heights.

### 3.3.4 – Line Spread function

This test was performed to check how and when the line sensor starts to detect the white line, and how height affects the detection as well. The test started by placing the stripboard on the testing rig, and slowly moving the piece of track by small intervals and record the voltage of the resistor that's connected to the phototransistor. The results of the test are shown below:

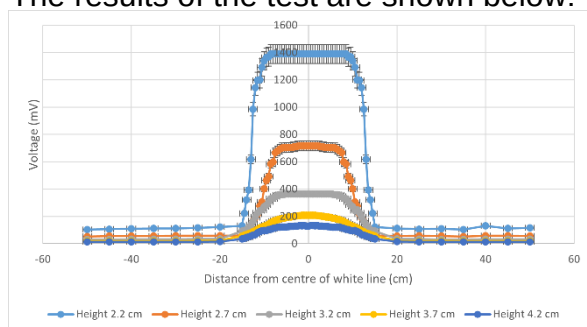


Figure 3.3.4.1: TCRT5000 LSF

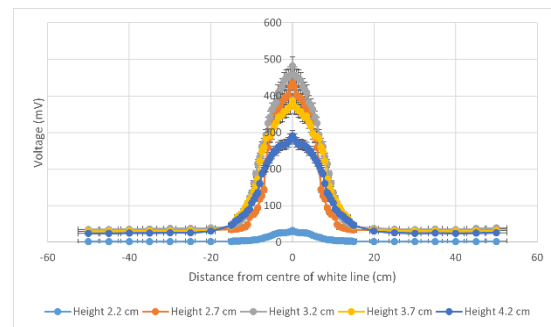


Figure 3.3.4.2: TSHG6400-TEKT5400S

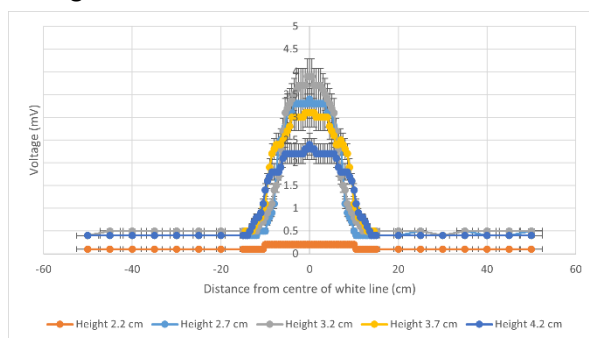


Figure 3.3.4.3: TSHA4401-BPW17N LSF

The following graphs above are shown how the sensors will cope with line breaks, as it can be clearly seen how the sensors react to the white line based on the steepness of the line graph. In Figures 3.3.4.2 and 3.3.4.3, the sensors appear to have a peak voltage when it is directly above the line, and these peaks are not as pronounced as the one in figure 3.3.4.1, and show a greater spread in

values, indicating that the detection of the white line will not be precise. However, for the TCRT5000 sensor, a clear, stable response appears when it goes over the white

line, indicating that it detects the white line even when it is not over its centre, which is the most practical option for real life situations. The TCRT also reads a much higher voltage than the two other sensors, which creates a distinct range that could be coded into the microcontroller to allow it to determine if the sensor is over the white line or not precisely.

### 3.4 – Errors in the graphs

The graphs that are based on the laboratory exercises include error bars. Error bars have been inserted to show the range of possible results based on the random errors that may occur in the measurement, such as inaccuracies in the DMM as it only measures to 6 decimal places, inaccurate power supply from the myDAQ as the voltage it gives out is around 5.1 V. Errors were put at 5% of the values measured in order to create an accurate estimation of how values may be affected.

### 3.5 – Conclusion

To conclude, of the three sensors that were created and tested, the TCRT5000 appears to be the most superior of them all, since it shows a great difference in the voltage it measures over white and black areas, giving it a very high sensitivity, as well as its ability to not get affected by external factors such as sunlight. In addition, the TCRT5000 has fixed geometry, since it is not handmade, which makes it more consistent in detecting lines, unlike the two other sensors that have been created. Due to these circumstances, the TCRT5000 has been chosen as the sensor that will allow the buggy to manoeuvre accurately on the track.

## 4. Circuit Diagram for Proposed Line Sensors

### 4.1 Full Sensor circuit diagrams:

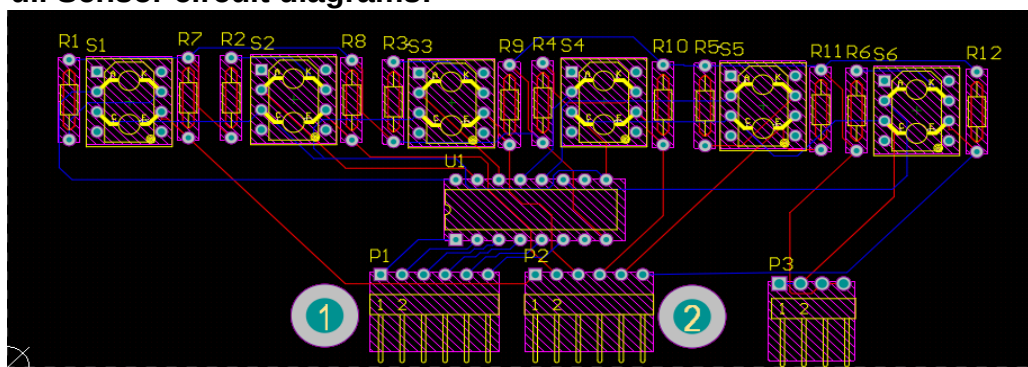


Figure 4.1.1 PCB layout diagram

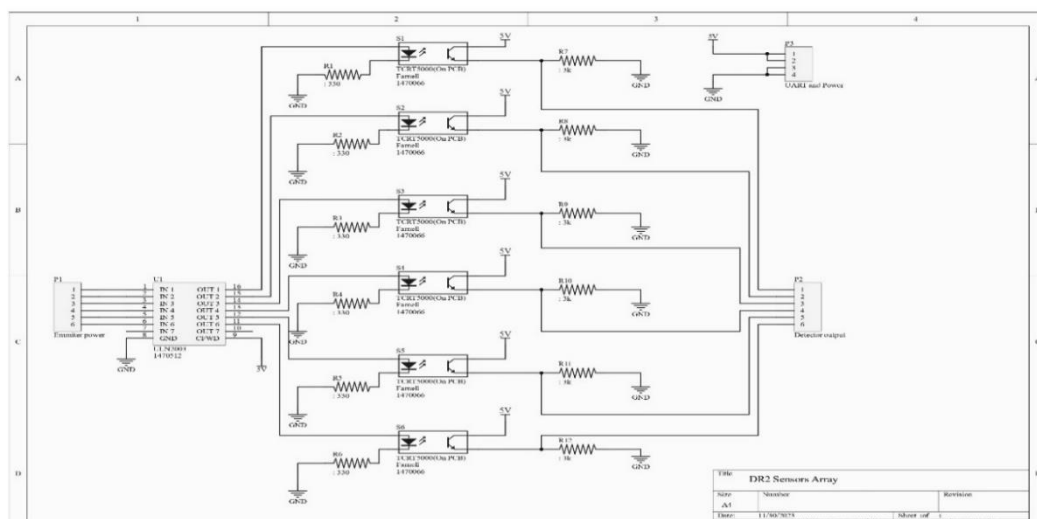


Figure 4.1.2 schematic diagram



## 4.2 Peripherals and Microcontroller Interface

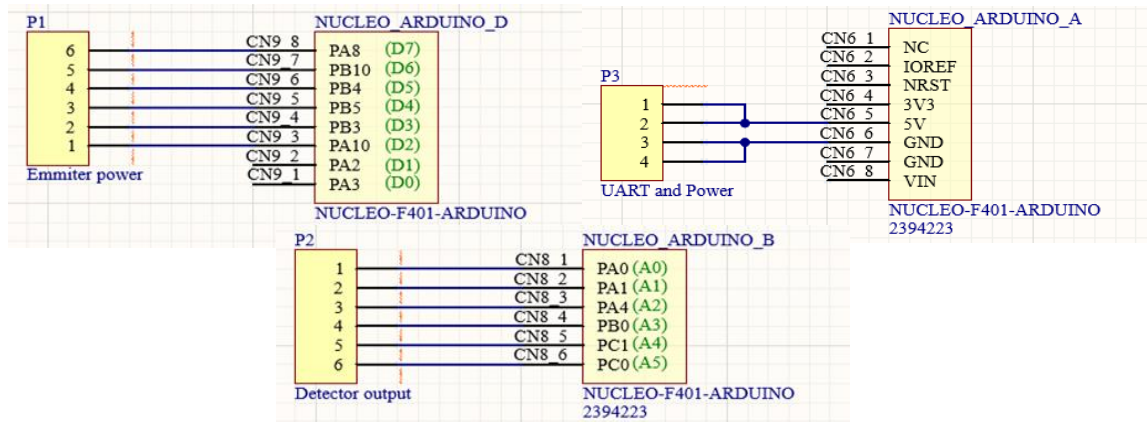


Figure 4.2 Wiring Diagram

## 4.3 The Discussion of the Wiring Diagram

The decision was made to employ three-pin headers for linking the board to the buggy. P1 corresponds to the emitter power on each sensor, thus linking to digital output pins on the STM32 (which are from pin D2 up to pin D7). P2 is designated for reading the output voltage of the sensors and is consequently connected to analogue pins of the STM32 (From pin A0 up to pin A5). Lastly, P3 serves for power supply to the board.

## 5. Non-Line Sensors

### 5.1 – Understanding non-line sensors

Various issues may arise in the way that the buggy will be controlled, for example, if the buggy is moving and it is required to then come to a complete stop, the motor can be turned off, which will take a lot of time to reach a speed of 0 m/s, or the microcontroller can reverse the current in the motor to bring it to a complete stop, however, how will the microcontroller know that the wheels have stopped moving. One way of solving this issue is by using speed sensing.

The advantages of using speed sensing include:

- Having the ability to stop quickly.
- Having the ability to follow the line closely and quickly.
- Being able to measure the length of the track.

Many different types of sensors can be used to achieve speed sensing, including encoders, motor current sensors, and battery monitors.

#### 5.1.1 – Encoders

The encoder is equipment used to measure rotation speed, direction, position, angle, and length. Encoders convert mechanical movement into electrical signals. Encoders are mainly used to estimate speed and position. The speed sensor includes an encoder disc on the motor shaft, working with a photo interrupter and an IR emitter-photodiode pair. As the light passes through slots and the solid disc, the optical sensor produces a fluctuating square wave output. The waveform's characteristics depend on sensor position, response time, and slot size.

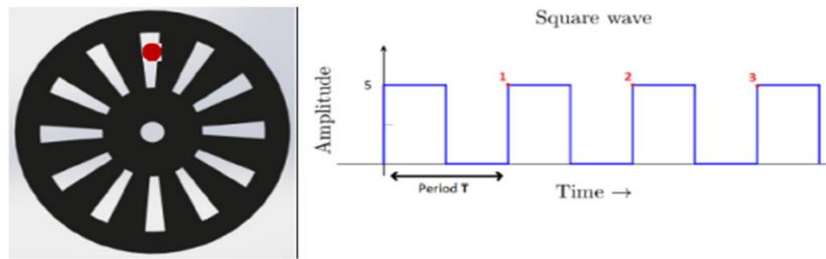


Figure 5.1.1 speed sensor and the output signal

The encoder shown above measures speed only, without telling you if the buggy goes forward or backwards. Quadrature encoders come into play here, as they measure both speed and direction. It can be interfaced with the microcontroller using GPIO interrupts with the incrementation and decrement of a counter variable.

The quadrature encoder will provide readings to the microcontroller to allow it to modify the PWM signals that are used to drive the motors. This will allow for the control the velocity of each wheel individually, enabling the steering of the buggy in the desired direction.

### 5.1.2 – Current sensing

Two isolated current circuits are already implemented on the motor drive board, and will be able to determine the current on the armature windings of the motor. These circuits will be calibrated to provide the microcontroller with a scale that ranges from 0 V to 5 V to represent the minimum and maximum current.

## 6. Control

### 6.1 How buggy direction is controlled

Within the buggy design, the two sensors S1 and S2, that have been strategically placed at the front, actively collect and relay data to the corresponding wheels M1 and M2. These sensors S1 and S2 measure the deviation of the buggy from the intended line, providing real-time feedback. Leveraging this information, the rotational speeds of M1 and M2 are adjusted accordingly, correcting the buggy back onto the designated path. For instance, in the event of a deviation to the left, the system modulates the rotational speed of the left wheel, accelerating it, while simultaneously slowing down the right wheel. A large deviation may mean a 50% power supply to the fast wheel and a 20% power supply to the slow wheel,

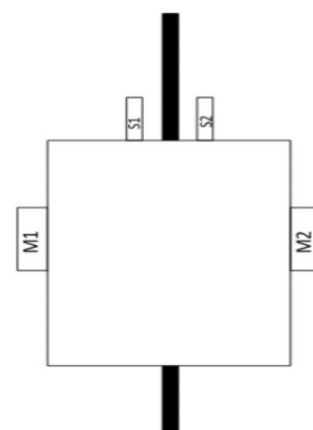


Figure 6.1

### 6.2 Proportional vs bang-bang controllers.

Proportional(P) and Bang-Bang controllers are two different types of control systems used in engineering and automation.

Using proportional control, the sensors S1 and S2 would each collect real-time readings of their distance from the line, and continuously compare these values with a 'reference value', which is a pre-set value that is an estimate of the distance



between the sensor and the line when the buggy is on course. The difference between the reading and reference value would be the 'error', providing data on the distance between the buggy and the line. An error of 0 would indicate the buggy is directly on top of the line.

Using Bang-Bang control, the sensors S1 and S2 would both give a measure of their distance away from the line. The difference in their feedback values would give an accurate estimate of the buggy's position relative to the line. A value of 0 would indicate the buggy is either directly on top of the line or has completely strayed from the line, while a large positive or negative value would indicate the vehicle is leaning on one side of the line. When the value given by S1 is larger than the value given by S2, it would mean the buggy is leaning on the left and vice versa.

Measurements from proportional controllers would often have a fixed error, as the reference value may be inaccurate, resulting in a steady state offset that leads to the buggy staying on one side of the line. This may prove problematic in situations where precise positioning is crucial.

Both types of control may be susceptible to oscillations around the desired path, especially when the system switches directions abruptly at high speeds. The on-and-off nature of the control system may cause the buggy to overshoot the line and then correct itself strongly, leading to a zigzagging motion.

Both types of control are also sensitive to sensor noise, possibly from beams of sunlight or variations in line spread. This may cause unnecessary fluctuations in the direction of the buggy, impacting overall stability and performance.

### 6.3 Brief description of PI and PID control

As mentioned above, proportional controllers are prone to steady state error. To eliminate this inaccuracy, integral action may be applied to the controller, resulting in PI control.

The output(U) of a PI controller is given by the sum of the proportional and integral terms:

$$U(t) = K_p \cdot e(t) + K_i \cdot \int_0^t e(\tau) d\tau \quad (1)$$

where U(t) is the control output at time t, e(t) is the error value at time t, and K<sub>p</sub> and K<sub>i</sub> are the proportional and integral gains respectively.

The advantages of PI control are that it is effective at eliminating steady state error which P control lacks, while being simpler to implement than PID control. However, PI control lacks versatility and may not be as responsive as PID control when experiencing sudden changes, for example line breaks or direct sunlight.

PID control adds another layer of stability on top of PI control by incorporating derivative gain, a constant which determines response to the rate of change of error. This helps to dampen oscillations and improve stability by smoothing out the control system's response.

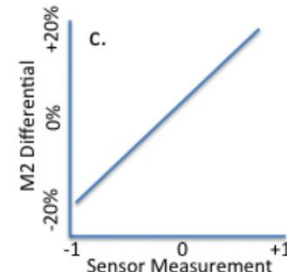
The output(U) of a PID controller is given by the sum of the proportional, integral and derivative terms:

$$U(t) = K_p \cdot e(t) + K_i \cdot \int_0^t e(\tau) d\tau + K_d \cdot \frac{de(t)}{dt} \quad (2)$$

Where K<sub>d</sub> is the derivative gain.

The advantages of PID control are that it provides the most versatility, stability and responsiveness compared to P and PI control. However, PID control also has the highest complexity out of the three control methods, due to it having three PID parameters that require fine tuning to achieve the optimal responsiveness and steady state error elimination. If the additional  $K_d$  parameter is not tuned correctly, the responsiveness of the system may be too slow, resulting in a buggy that is unable to correct itself in time before it strays from the line. Derivative action may also amplify any noise from the sensor measurements, leading to further inaccuracy.

The rotational speed of each wheel would be the most important factor to control, and it would be determined by the output of the PID controller,  $U(t)$ . The output would indicate how far away the buggy is from the centre of the line, with -1 being far to the left and 1 being far to the right. Depending on the direction and severity, a difference in wheel



Graph 6.3

speed will be implemented to correct the buggy back onto the right track.

#### 6.4 Relationship of chosen algorithm to sensor design

After considering the characteristics of different control algorithms, PID control was chosen, as it offers the most flexibility and pairs well with the TCRT5000 sensors on the buggy. The sensors provide a measure of distance between the sensor and the line. By comparing the difference between two symmetrically arranged sensors, the error value can be processed by the PID algorithm to generate a real-time output.

#### 6.5 Sensor implementation, number geometry and interface to the microcontroller.

TCRT5000: 6 sensors, located at the front of the buggy 18 mm off the ground. Connected to the pins A0, A1, A2, A3, A4, A5 respectively.

Encoder: 2 sensors, attached to the wheel axes in the gearbox. Connected to the D12, D13 pins respectively.

Current sensor: 2 sensors, located on the motor drive board. Connected to PC\_2, PC\_3 respectively.

Battery monitor: 1 sensor. Connected to PC\_5.

#### 6.6 Coping with line breaks and direct sunlight

Line breaks and direct sunlight are practicalities that may cause the buggy to stray from the designated path and must be overcome.

Whether it is using step control or proportional control, line breaks should not be a problem. This is due to the nature of the way that the sensors relay feedback to the system. The sensors return the same value (0) when the buggy is directly on top of the line and when the buggy has completely strayed from the line. This value would tell the system that no correction is needed, meaning the buggy will continue in its original direction until it reaches the line again. However, in practice steady state error cannot be fully eliminated. In the event of a line-break, this error would cause a constant deviation, so the buggy would attempt to correct itself according to the error, causing it to stray from its original course. In this case, derivative action would be the solution to dampen the system's response, preventing overshooting and oscillations before it reaches the next section of line.

Direct sunlight on the sensors may create intense contrasts and shadows, causing fluctuations in readings and unnecessary changes in direction. Integral action may help to eliminate error caused by sustained exposure to sunlight but does not cater to rapid changes in lighting. Derivative action will therefore be the primary method with which the system maintains stability through direct beams of light. A more direct and effective approach may be to glue plastic shields around the sensor to avoid direct sunlight exposure, while also minimizing sensor noise through limiting background light.

## 7. Hardware Overview

### 7.1 The Chassis and Stress Analysis

To design the chassis, the most important thing to consider is the deflection between two gearboxes and wheels that are under the back of chassis. Due the castor and the wheels support the whole buggy, and the weight of buggy is mostly at the center of mass, so the chassis may appear the slight down bending. The Equation (3), where  $\delta$  is deflection,  $W$  is the load,  $L$  is the length of the chassis,  $E$  is Young's Modulus, and  $I$  is the area of moment of inertia, Equation (4), where  $b$  is the width,  $h$  is the thickness of the board, Figure 7.1.1<sup>[4]</sup> can be shown that the two fixed beams supported a board that can distributed the load, Figure 7.1.2 is the simply supported board

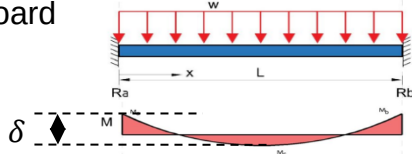


Figure 7.1.1<sup>[4]</sup>

$$\delta = \frac{WL^4}{384EI} \quad (3)$$

$$I = \frac{bh^3}{12} \quad (4)$$

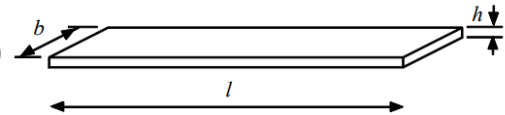


Figure 7.1.2<sup>[2]</sup>

Assuming that using the Acetyl material, according to the procedure handbook<sup>[2]</sup>, get the  $E$  is  $2.8 \times 10^9$  Pa. For the chassis design, get the  $b$  is 0.13 m and  $h$  is 0.003 m. Using Equation (4),  $I$  is calculated  $2.925 \times 10^{-10} \text{m}^4$ . By adding all the buggy components, the  $W$  is  $66.5 \text{ Nm}^{-1}$ ,  $L$  is 0.23 m. Using Equation (3), the deflection  $\delta$  is calculated 0.592 mm. Because of the deflection is too small, it cannot occur obvious bending, so can ignored the deflection issues.

The SOLIDWORKS simulation can do the stress analysis. It can clearly observe the stress contribution for the whole chassis under the different materials that can choose the best material which applying in the chassis. For the stress simulation in SOILDWORKS that is used the Von Mises stress.

“The Von Mises stress is used to evaluate the strength and durability of a material when it's subjected to external forces.”<sup>[4]</sup> According to the result in SOLIDWORKS that be shown in Figure 7.1.3, Figure 7.1.4. and Figure 7.1.5 below, by comparison of Acetyl, Aluminum and Mild Steel material that applied in chassis, assuming that the applied distributed load is both 15 N, compare with these two diagrams, it is obviously found that the deformation scale and the maximum value of Von Mises Stress in chassis by using Acetyl is evenly smaller than using Aluminum 7079 Alloy and Carbon Steel. So, choosing the Acetyl material may not have obvious deformation that cannot especially affect performance.

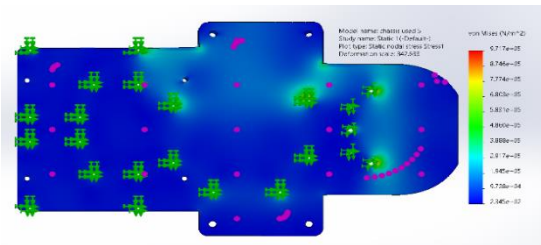


Figure 7.1.3 – Acetyl Stress

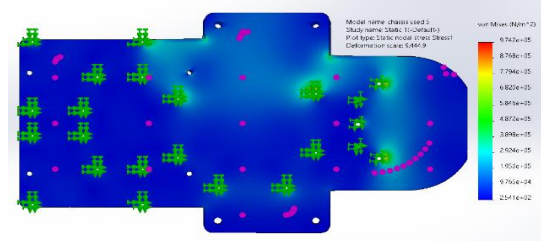


Figure 7.1.4 – Aluminium Stress

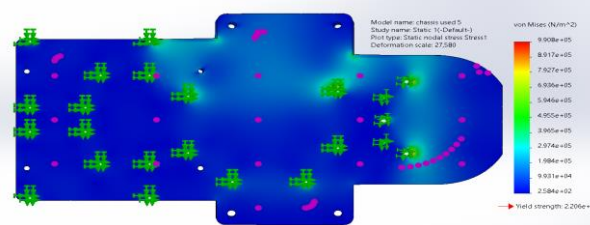


Figure 7.1.5 – Mild Steel Stress

## 7.2 The Material Choice and Cost of Manufacturing Comparison

The Procedure Handbook shows that there are four Material can be chosen, The Table 7.2 is shown below that the comparison of these four Material and can be decided to choose the best one.

Table 7.2 – The Comparison of four materials

|                                  | Acetyl | Aluminum | Glass-reinforced Laminate | Mild Steel |
|----------------------------------|--------|----------|---------------------------|------------|
| Density (g cm <sup>-3</sup> )    | 1.8    | 2.7      | 1.9                       | 7.8        |
| Young's Modulus (GPa)            | 2.8    | 69       | 11.5                      | 207        |
| Cost (£ / 0.5 m x 0.5 m x 3mm)   | 29.47  | 21.52    | 60                        | 27.68      |
| Power used in Laser-Cutting* (W) | 40     | 3800     | 250                       | 500        |

\* Data from the 3.2 mm thickness sheets <sup>[5]</sup>

According to the Table 7.2, the smallest density is Acetyl sheets that could buggy lighter, the Young's Modulus is also the smallest, that could the chassis has high flexibility and the whole buggy has effective damping and the benefit of lower Power used in Laser-Cutting is that the laser can cut the chassis that is made of Acetyl easier than the other materials. In addition, the price of copper is slightly higher than that of other materials, but its price is still average. Overall, choosing copper is a cost-effective choice.

## 7.3 The Weight, Wheel Drive and Layout

1.For the estimated weight of buggy, the chassis and the battery box are estimated as 500 g, and adding Microcontroller board and its breakout board, Motor drive board, Battery pack, Motors + gearboxes + wheels + encoders, Ball castor. Those can be found that the weight via ESP Technical Handbook <sup>[1]</sup>. The total estimated weight is 1342g, the maximum weight can expected be limited at 1500g (approximately 15N).

2.For the wheel drive, the assumption is that the Ball castor put on the front because it can be turning flexibility, the stress distributes the pressure to the front, so it has the

high drifting while the buggy running.

3. For the layout of the buggy, the overall chassis design is streamlined, because it can reduce the air friction. The battery pack should be installed under the chassis to balance the center of mass. The layout of Sensor PCB will be located under the chassis is that the sensor can control to following the white line more stable due to the enough height between the ground and the PCB, it is proved from Chapter 3 that the maximum is 35 mm, and the minimum is 22 mm. For Figure 7.3, the distance between the center of castor ball is 19.24 mm and the radius of castor ball is 9 mm, so the height is 28.24 mm that is standards compliant.

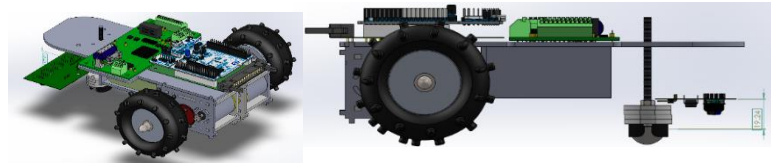


Figure 7.3

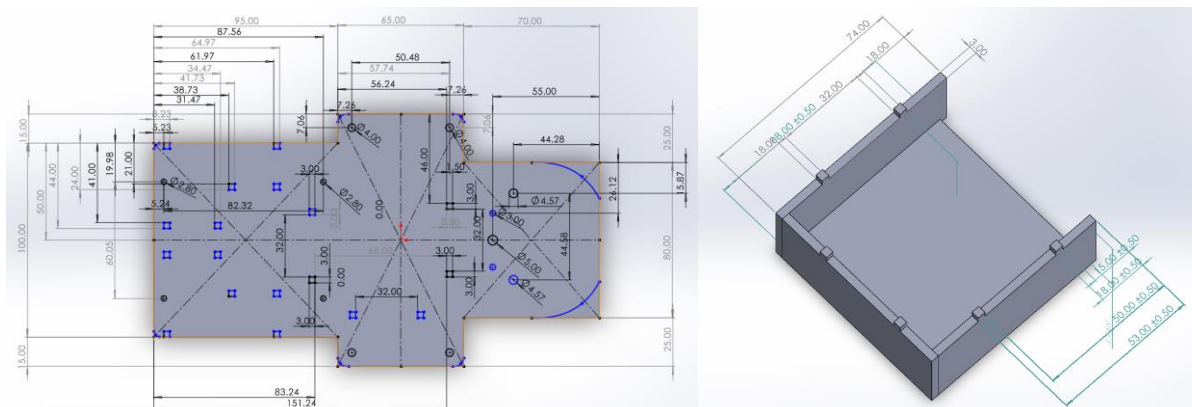


Figure 7.4.1 – 2D Drawing of Chassis and battery

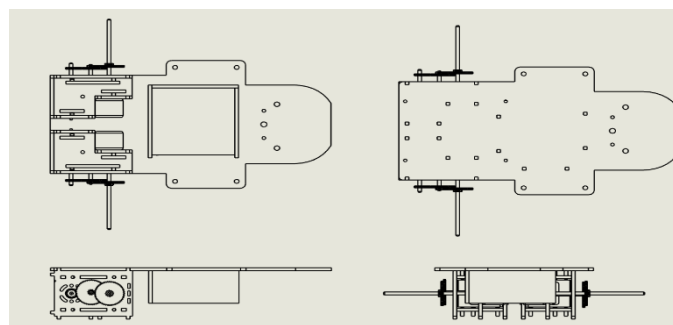


Figure 7.4.2 – 2D Drawing of Chassis with Gearboxes and battery box

The Figure 7.4.1 and Figure 7.4.2 above show that the 2D Drawing of Chassis and the details of design of the buggy. Consider the sensor PCB which put near the ball castor to control the direction that can follow the white line, so, on the front of the chassis that reserved the space to add the sensor PCB in the future. The reason of there is an open mouth on the left side of the buggy is that it can be convenient to change the battery when the battery has no charge. These designs are not fixed and may be modified in the future as circumstances change.

## 7.5 Other Hardware thoughts



During the design, the axle that connect the wheel should be longer, because if the axle is short, the wheel hub may touch the gear that can affect the buggy running. The distance between the two motors is also significant, the minimum track width is 0.3 m, to have the best turning angle, the total width of the buggy should not exceed 0.2 m, so the distance between the two motors should not be too big, but should not be too small, because when two motors are working, it may cause mutual influence between the two motors.

## 8. Summary

The aims of this report were to select the optimal sensor for the buggy, which is capable of remaining reliable in various circumstances, explaining how this choice informs our chassis design and finally elucidating how each element in the overall system integrates with one and other in order to create a closed feedback loop.

A height of 28 mm has been chosen for the sensor bracket based on our results from the sensor tests. The difference in voltage of the phototransistors resistor when it is above the white line versus when it is above the black line at a height of 28 mm is around 500 mV, whereas for the TSHG6400-TEKT5400S its around 400 mV and for the TSHA4401-BPW17N, its 2.2 mV. The largest difference in voltage gives the sensor of the highest sensitivity, and hence the TCRT5000 is the most sensitive. A height of 28 mm, the TCRT5000 has a clear detection view of the white line, as in when it is above any section of the white line, a stable voltage of around 700 mV is measured, whereas the other two sensors have unstable voltage measurements when they are above different sections of the white line.

The TCRT5000 displayed the most significant voltage difference (500mV) when positioned over a black line versus a white line at a height of 28mm from the ground. In comparison, other options like the TSHA4401 (2.2mV difference) and TSHG6400 (400mV difference) showed smaller distinctions at the same height, this comparison accentuates the TCRT5000 capabilities of line isolation and identification.

In conclusion, the TCRT5000 was selected due to the aforementioned calculations.

## 9. References

- [1] ESP Technical Handbook, The University of Manchester (2023). Available: [https://online.manchester.ac.uk/webapps/blackboard/execute/content/file?cmd=view&content\\_id=15044517\\_1&course\\_id=78308\\_1&launch\\_in\\_new=true](https://online.manchester.ac.uk/webapps/blackboard/execute/content/file?cmd=view&content_id=15044517_1&course_id=78308_1&launch_in_new=true)
- [2] ESP Procedures Handbook, The University of Manchester (2023). Available: [https://online.manchester.ac.uk/webapps/blackboard/execute/content/file?cmd=view&content\\_id=15044516\\_1&course\\_id=78308\\_1&launch\\_in\\_new=true](https://online.manchester.ac.uk/webapps/blackboard/execute/content/file?cmd=view&content_id=15044516_1&course_id=78308_1&launch_in_new=true)
- [3] 11 Beam Deflection (date unavailable), Civil's Guide [Online]. Available: <https://civilsguide.com/beam-deflection-formula-tables/>
- [4] "What Is Von Mises Stress?" , Simscale [Online]. Available: <https://www.simscale.com/docs/simwiki/fea-finite-element-analysis/what-is-von-mises-stress/>
- [5] Manufacturing Processes Reference Guide (1994), Robert H. Todd, Dell K. Allen, Leo Alting, Industrial Press Inc., P188, Available: [ISBN 0-8311-3049-0](#).